FULL STACK DEVELOPMENT – DAY 3 SUMMARY

TODAY COVERED TOPIC:

- COMPILER AND INTERPRETER
- FEATURE OF JAVA
- JVM(JAVA VIRTUAL MACHINE)
- JRE(JAVA RUNTIME ENVIRONMENT)
- JDK(JAVA DEVELOPMENT KIT)
- IDE(INTEGRATED DEVELOPMENT ENVIRONMENT)
- MAIN FEATURES
- JAVA'S MAGIC BYTECODE

- **COMPILER AND INTERPRETER:**

  COMPILER:

  Compiler is to find a syntax error and source code (.java file) to convert bytecode(.class file) .

  INTERPRETER:

  JVM is use the principle" WRITE ONCE RUN ANYWHERE". Interpreter use JIT for speed and its convert byte code into machine code. Its auto check error in line by line .

- **JVM(JAVA VIRTUAL MACHINE**):
  - JVM is a important to run Java Programs.
  - JVM convert bytecode into machine code

  JVM Does(work):

  - Loads Code
  - Verifies Code
  - Execute Code – converts Bytecode into Machine Code using:
    - ✓ Interpreter
    - ✓ JIT(Just In Time compiler)
  - Manage Memory

- **JRE(JAVA RUNTIME ENVIRONMENT):**
  - It Include JVM & Libraries(pre – built code).
  - JVM is using Javac compiler to compile the program
  - Inside JRE:
    - ✓ JVM
    - ✓ Core Libraries(java.util, java.io,java.lang)
    - ✓ Supporting files
  - Example:

    Run Java program but not write or compile programs

- **JDK(JAVA DEVELOPMENT KIT)**
  - JDK include Write, compile, debug , and run java programs
  - Its Inside :
    - ✓ JRE

- ✓ JVM+ Core Libraries

- Development Tools:
  - ✓ Compiler (javac)
  - ✓ Debugger (jdb)
  - ✓ Documentation tool (Javadoc)

- **INTEGRATED DEVELOPMENT ENVIRONMENT:**
  - Popular IDEs for Java:
    - ❖ Eclipse
    - ❖ IntelliJ IDEA
    - ❖ NetBeans
    - ❖ VS Code
  - Programmer's Workspace.
  - Complete environment to write , edit, compile, debug, and run code easily.
- **MAIN FEATURES:**
  1. Code Editor
  2. Compile/Interpreter Integration
  3. Debugger
  4. Project Management
  5. Other Tools (build tools, version control(Git), Plugin Support, Testing frameworks)

- **JAVA'S MAGIC BYTECODE:**
  - If compile a program , becomes a .class file (bytecode)
  - Every .class file starts with special code.
  - Secret Signature start of every java .class file. This is called Magic number.
  - Its like a ID card of bytecode.
  - When JVM opens a .class file first thing it checks whether it starts with special code.
  - If its there – JVM Knows it's a valid Java bytecode files.
  - If not – JVM rejects it (Prevents error).